There's plenty of room in the front Please take your seat slowly welcome everyone we start There by now they used to this kind of having a survey at the beginning of a quarter, right? We start with our quality assurance team having Something to ask you for I don't know exactly what but they they have this Okay Hello, it's very good to see you are still here after a week three quarters of Education and as usual I have some questions for you, but first let's do some Yeah, random checking so you had a couple of courses last quarter And I want to know which one you think is the best one who thinks that electronic circuits was the best course in Q3 Nice nice good to see And for the EE students who thinks computer architecture was the best course Nice Probably something that's only for the dedicated programmers maybe Who really liked physics for EE? some physics lovers Automotive students who thinks spectrum of automotive was the best course they had in Q3 Okay, good and Who thinks physics for AT was the best one? Okay, I see some hands it's good to see that all Courses were pleased like by some people but of course I would like a bit more detailed Information from you, so please if you can go to this website and fill out the survey That will be incredibly helpful for us Maybe oh that's can also say some words as a teacher why evaluation is important Or what did you change for this year based on the feedback of last year? Yeah, that's actually a very good question Yeah, we we we often don't get enough feedback, that's our biggest problem and With with the problem with not getting enough feedback is that we don't really know what the average student thinks what we tend to see is Split distribution between students who for what reason like the course And they have something nice to say and students who for whatever reason did not like the course and say have something to say and the average students who just said it was okay, they don't bother filling the survey and Statistics just says that the average group of students about 60% of 70% of students So we get very little information what we can do, but when we get information we use it So first of all you have the student body I don't know if anybody in this room actually is participating actively student body But if some of you are part of student body, you know that you have meetings regularly during the quarter to give feedback So you are you are collecting information and this feedback is fed directly to us So for example previous year students tell us that the instruction moments were not effective for them They said it was too much of a waste of time. So this year we're trying to again change how we organize the instructions so The moments where we are here already because these are scheduled. We have eight hours a week, which are scheduled the moments We are here we can use them to your best the best purpose that you think it there is for you What you get the maximum out of it We created a lot of quizzes online for this course a lot And we know the students don't really use them or don't use them in the right way So you want to work with you to try to work with these quizzes? exactly work with these with these quizzes as effective as possible for you because Science So not me science tells us that the best way to learn something is to test your knowledge again and again and again So these are just two example. There are more of those but they are Thank you very much or that the survey will be opened for two weeks.

It will close on midnight on the 5th of May Please complete the survey.

It's extremely valuable for us. Thank you So let's let's try to give you five minutes more to complete the survey, then we get can get started We can get this show on the road Yes, any anyone is still very busy answering the survey needs a couple more minutes, yes, okay, please go on it's valuable If you haven't started, please copy the take a picture of the of the web address or something on the board see Thanks Yeah, yes, it's a very nice job by the way Signals Well Okay, I think I hope it works now While that is still writing And yes, I'm done.

Thanks Well, that is still writing. I just wanted to give you the feedback that This list is actually part of the feedback.

So just that you know that we take feedback quite serious We learned in the past that because what that will say more about it in a second But we prefer to work with the with chalk and the chalkboard And we heard it can sometimes be a bit confusing because we don't have slides There's no structure and so on and so forth So we now have a topic list that we always put at the beginning of a lecture on the board. Yes used as guidance True. Thank you. Thank you for saying that So welcome.

Welcome to the communication one course my name is Oded Raz and Mark Spielberg is my co-teacher We will take you in the next eight weeks nine weeks actually because there are all sorts of intermediate weeks through the basic questions topics which are related to To communication and I put online So first I want to share this slide which is important to us The University of Ainuvan and specifically the faculty of lecture engineering Cares about your well-being and wants to promote it in any possible way There are multiple ways for you to engage with the faculty to get support if you need it Well-being has multiple components to it could be financial. It could be personal. It could be health related You might be suffering from whatever disability there is you might become if our country which is currently being in a war all of this can affect your well-being and struggling with whatever you are struggling Feel free to contact any of the academic advisors we have You can also come to me if it helps if it if you feel that you have a face that you know or to mark We're always happy to try to help whatever we can to do to help you because well-being comes ahead of learning the course I am assuming you realize that's very true in life in general. So that's really important Then we were asked by By else to to tell you what we are changing and we are trying to change and this year is another Twist small twist, but we were constantly doing this I mentioned when I when when else asked me to say something about Studying I mentioned that One of our biggest challenges is to get you to understand the course material. This is not a trivial Course, although some of you may find it trivial, which is very nice for you. I'm very happy for you but many students struggle not because They are not capable But because for whatever reason either we are not doing our job correctly or you're not using what we provide you in the right way These are two options and there may be other options that are there The only thing that we care about the only thing care about as teachers is that you come out of the course understanding something new I'm being able to use that new knowledge for the rest of your studies We are here to let you pass the exam and pass every successive assessment in the course because that's what we are Basically paid to do right? This is part of our yeah, so we want you to succeed even if sometimes you say but the exam was very tricky or The teacher tried to trick us. We're not trying to trick you. This is not the purpose of what we do We're trying to test if you understood something and in order to understand and that I spent a lot of time trying to figure out What to do the only thing that really works really really works have been tested again again in studies is to test your knowledge Now testing your knowledge and that's very important in the age of LLMs testing your knowledge is not to take the question from the quiz give it to chat GPT and get the answer and then read The answer and now I understand Because reading the answer that the chat has given you constitute no understanding Because you won't be able to apply the knowledge for the next question and the next question Most likely will be in an exam where you don't have chat GPT available For now, maybe there's a future where it's there, but for now you don't have it We have two assessment moments in this course one on the 18th of May to be after hours We need to schedule is it five thirty or six? I think it would most like to be six because if five thirty is too short between Well, we'll have exam written exam multiple-choice written exam in Probably will take over the entire auditorium because we have 500 472 students in the course, so we need to fit all of them and This midterm exam is good for 20 can be good for 20% of your grade But if you dug deeper into the study guide into Osiris you saw that if you do poorly on the midterm Which can happen? You can compensate by doing better on the final exam But we encourage you so that means technically you can decide not to show but we encourage you with all possible Encouragement to come to the midterm Even if you haven't learned haven't prepared you had a bad day Whatever Come test your knowledge get exposed to the experience of taking exam in this course because the final exam Will have elements in it which will look like the midterm exam Right will have multiple choice. Yeah, and I think what is what is a bit of I don't know That's feedback that we get every year is that our exam at the end of the course is never the same So that's our biggest complaint and we are proud to say so that we achieve it every year so that sounds a bit strange but We are not here to let you exercise all the exams of the past year and by default You will know how to do the exam now We try to do every year check your knowledge because or not much your that your understanding of the course material That's what it is about that you understand What is the course about and the exam is passable if you engage with the course and the course material? But if you just try to do past exams and then thereby try to pass the course that will be difficult However, the midterm is a really good possibility to learn from our okay, it's my first interaction with how a debt and mark are setting up a midterm or Preparation for the exam so I can test myself you have nothing to lose as it had already mentioned if you do poorly in a midterm Okay, then you learned that and you know what you have to turn or you work on for the final exam But it's definitely a moment that you should not Ignore that easily.

Yeah Just one example. We are also quite open with all the exams. I think we have the last 20 iterations online Yeah, be my guest do them all It will probably not help. It's better to No, seriously, it's actually better to try to be engaged with the course material ask the questions in the contact moments that you have Well, don't discourage him completely It does of course help to exercise and there will be a lot of Exercises material in this course at some point students complain. There's too many options That's something we talked about before Students one of the feedbacks we got from students. There are too many things to do I'll give an example after this lecture is over. It will appear as a video lecture in Something there's something there.

I don't every every three years. It's a different name. I'm sorry It will appear Which means that now some of you may ask yourself should I watch the lecture again? But then you'll find that in the same at least on my screen and I assume you get the same There's also at the same lecture from last year and from two years ago Now I promise you what I told students to eat last year and what last year was a mistake because the audio went out halfway, but okay, but I told you two years ago is In some parts different what I'm telling tell you today because I'm ad-libbing completely I know what I want to tell you, but there's no there are going to be slides This is one of the last slides you see for me by the way. I'm not using slides at all so You might ask us if should I watch the video from two years ago the answer is no if you are here If you're here You've consumed the content for that module for that topic for this year for this course if you make your own notes you have the stuff you need to continue learning this course if You feel that you miss something you want to watch it again. That's why we have video content of course available There's a reader for the course so we don't provide PowerPoint style. There's a reader. It's a very expensive 150 pages of reader You can read it to prepare for the lecture. You can read it after a lecture. It's up to you what you do with it Last year I made podcasts for students Seriously, I don't have still on the website, so I didn't make them myself. I used a Google notebook LM I don't know if you know Google notebook notebook, and you know this so that you put a PDF and you get a podcast So there's for every chapter in the book. There's a 10 to 15 minutes podcast two people talking about the course American accent very have American accent nothing wrong with it. Just sharing I found it fantastic If you're cycling to the court to the lecturer you put on this on the on your you can study while cycling. It's good So there is a lot Don't try to do everything That's the message Don't let yourself be overwhelmed the quantity do what you need and if you're not sure ask If you say I watch I saw there's a lecture do I need to know this or do you need to watch it as well? It's okay. We are here to answer these questions. There's a discord channel, which I showed before you can use that to ask questions So you can you can you can link to discord you can ask question you can share questions It's all there okay, I was talking about the course so we have a midterm The course has seven active weeks of learning and a one-week repetition That's the start or in the electrical engineer at the University in general rights and but because of Vacation days and also the celebrations we have nine weeks of the course The study guide tells you which week is happening what and there are three weeks in this course where there are labs We'd like to say more about the lab's work Okay Yeah, so it's actually four weeks, but it's three because again of these separation, so we normally have During a week we have two contact moments or Tuesday afternoon Thursday morning, and we use these same moments for Doing the labs the labs are offered for you to be present in person, but you don't have to you can also do them at home While you're present we encourage you to even work together with others So it is something that we intend for you to work together and try to figure out how it works And at the end of the lab or if the lab we could get a Chris and this Chris is graded so they are in total three quizzes and these three quizzes they will happen on canvas and Based on these three quizzes you get a great and that counts for your final grade for this course So it isn't a component is of importance, and I said that we normally have two contact moments during a week But I think in the third week where the first lab would happen. Yeah one of the days Is that there's a national holiday, so then we have the Thursday I think and the week after the Thursday is ready as well. I think that's how it is that's separation. Yeah That's that's mainly that in these Lab sessions we also have 30 a's supporting you so they will support you with you asking questions It's really for you to learn what you're doing and all the content that you learn there You will probably need to answer the lab Great Chris great at Chris, so yeah indeed the quiz for the lab is based on what you've experienced in the lab So while in general kind of skills are useless because you just touch a plea to answer those here At least you need to do some work yourself. Yes Yes, okay So about quizzes and about quizzing in general so last year.

We decided to to help you with this Resistance to testing yourself by enforcing testing in the classroom We figure out if we'll give you a canvas quiz during the classroom We'll give you the time to answer it Then we can analyze together the results and look at where things went wrong in answering you will have a learning moment the result was that every student all the students went away and They didn't feel the quiz afterwards at home So you had no learning moment now it was waste of time for everybody Especially students get annoyed with us because they I'm sitting here for 45 minutes and nothing happens, so This is not gonna happen again. We learn from mistakes the quizzes that we use last year's for you to test your knowledge are now gonna be included in the in The modules as what to do for the next Lecture so we try to break down the modules in canvas, and you will see what to do in today's Tuesday So after the lecture today, there'll be a module say what to do for Thursday And what to do further is to take the quiz in this case about DB and DBM I think how to use decibels the first lecture yeah the first lecture so and then what we will do at the beginning of every lecture We will give you a short quiz just to see How where what you are it will be five ten minutes, so you can walk in the canvas quiz will be available You can do it then we can look at the results and We can see if there are big gaps from last lecture that we need to fill in that's one thing Well, and it will work better The more of you who actually use these Tools these quizzes the better. We know where you are the better. We can fit what we do here to what you need, so It's for you, but it's also for us What we do and you have two teachers, huh? This is very luxurious course have two teachers We're investing in you and the way we are going to interact is is is on average the following on the normal situation Most of the time I will give the lecture and I will spend today I think one half hours maybe a bit more depending on the clock to explain what I want to say today and We'll have some instructions We will try in That'll be my part Afterwards mark in most of these lectures will try to solve a question from old exams. That's true Or I asked you beforehand what you would like to do So we will we will try to make it useful for Thursday. I probably will ask you what you want to do In the in the court discord in this court. Yeah, okay. Yeah So we also value direct interaction If I don't get sufficient interaction, then I might propose an exam question from last year Just one small hint already So even after two lectures you will be already able to solve an exam question. So that's that's how that goes And we we try to keep you engaged in the instant But if you have the impression that this is not working for you also, let us know so we are quite adaptive to Feedback. Yeah, and because we have two teachers what we have decided to do is that if I will give the main lecture Then mark will start every session we meet by doing a recap of the previous lecture and He can use whatever he learned from your online quiz at home or we will do it interactively doesn't matter But he will also just go to the main points that we discussed last time So you'll have a refresher at the beginning so you're not landing from nowhere So if you miss something and you can ask questions The intention of this recap is not to quickly run for everything so we can move forward We feel that we have enough time in the way this course is organized in the number of hours We have available to teach that we can take our time. We're not rushing through anything If you want us if you have questions, please ask them. We encourage you to ask questions. Yes. There's a question. Yes Does it mean that we may be extend beyond the first two hours and there is it has happened in the past? Yes, it's so yeah, that's actually it was fake of course But yeah, it's for your information It's a disclaimer based on the fact that we are so creative with the starting of the lecture it might be that we need more time so that we actually go into the third hour to finish the content that we need to Finish because we have quite a clear schedule on what we want to achieve. Yes at what day of the the good thing about This course is we are not gonna be Migrating between the lecture and in structure as you've seen in my timetable where it's called We have four hour blocks here and on Thursday we meet in Atlas in the big lecture room in Atlas for four hours You don't have to move around you can just stay in the same place and enjoy the instruction part as much as you enjoy The lecture part or not depending on you Anything else the mist if not, I will let the PowerPoint presentation Which I will switch off in a minute is available online for you to use So yeah, let's miss which of our point.

Oh, no, I need this. Sorry. I'm still All right It's time to start So this course it's called communication one Call communication one for a reason This course will give you if more of you want to grab a seat I'll take that one minute of disturbance if you want to find a place to sit, please do that now squeeze in please It's okay, not a problem.

There is a place in the front. Yeah, it's very close very good All right, so This this course is a communication course And if this course would have been taught Even 20 years ago there will be a large part of this course will be if trying to answer the question How do you do communication for analog communication? Almost nothing is analog anymore.

So it's not a judgment. It's just a observation Almost all communication nowadays is done digitally And digitally doesn't mean that we just sends ones and zero, but it means that we are actually taking information which surprise surprise all information started somewhere analog Take a minute to think about it. You realize I'm correct so all of them are started some point analog, but we are consuming it in most cases digitally and The very basic question for communication is how do we take something or what do we need to be able to do? Maybe that's a better question. I'm Accepting all answers from the audience. What do we what are we supposed to be able to do if I have in one side a microphone? Picking up audio waves changes in perturbation of air density of air and the other side maybe the other side of the Building the country the continent a loudspeaker which supposed to emit these audio waves What do I need to be able to do? If I want to send this information somehow digitally from one place to another, please. Yes Okay, that's already jumping in.

Okay sampling the analog incoming signal. Why do we need to sample the information? Okay So clearly one of the things first thing to do because the signal is analog you want to convert to something which is digitizable And if you want to digitize something The first thing is you need to capture not the entire because the analog waveform is continuous in time Yeah, that doesn't surprise you if you want to send some information you need to decide I'm not gonna send the entire waveform because that will be analog again I'll try to capture moments in time of That analog information which I can then use to send the information You've done the course signals and systems, I guess so, you know that sampling is not something you just do There are criteria for sampling Yes, you all heard about Nyquist criteria We'll we'll we'll share an extra tear and maybe we'll explain why it's so important in a different way But definitely so sampling is important next. What else do we need to be able to do? ideas So I took the analog waveform I sample it now. I have a bunch of samples maybe yes, please Send to the other party.

Okay, so we need to but is are the samples good enough? So I will draw it so I'll make it clear for you. So I will take a waveform and As our colleague has told us we will sample it I'm trying to draw them equally distance because that's usually how we sample. So this is the time domain and I've sampled it. So now I have not Infinite values I have one two three four five six seven points.

Is that digital? Does it look digital? What is digital?

Yes, please It's discrete, but it's not digital right now because this could be 1.3 volts and this could be one point one one point two five one three five Nothing digital about it. So not something needs to happen yet. What do you do with this information? Yes Okay encoding.

That's a nice word to use. So we need to take this voltage Which is still an analog piece of information and somehow convert it to Bits, yeah, we know digital information is send its bits is stored as bits and this bit So we need a process where we take these different points and convert them into bits This process is is important.

It's called quantization. I mean I'll share the topic map for this course you will see all of it and Once you have gone from Discrete analog values to a binary representation. You now have bits which is good It's a good start and then you said I need to send them Okay What what ways you think about sending information?

How can you send information? With a wire You can wirelessly you can use a glass fiber for example if you want Yes, you can do that. Most of you have internet somewhere with the flower fiber connection. Actually all the data is connected with fiber Yes, we need to figure out how what happens when you take the bits Which we agree this we need to generate from these samples some bits We take the bits and we need to send them so we need to put them on the fiber or in the radio wave Somehow and send it to the other side. So we need to understand how to Convert the bits into something we can send we call this Symbols we need to convince and then we need to send the information And we need to detect it as well Okay, so that Oh Wonderful. Thank you. Yes, exactly. So If you send information there is a risk which is nonzero That someone information that arrives at the other side is not the information you sent happens all the time These are called errors bit errors.

They really happen all the time and some ways of sending information are more prone to errors than others It's it's the nature of how we're sending them this course We'll talk a bit about what happens to signals when they're sent but not in all details There are a lot of details which will be covered in other courses.

But if some of the bits which arrive are wrong an Example we are clever enough to convert the voltages into binary digits. It's okay. So I will decide that this This voltage is equal to one zero one one.

It's a number. It's a binary number if I send this sequence and I detect one one

one one because the second bit has flipped if can happen and then I try to Reconstruct the waveform we'll talk about the construction later. The value I will get is this Because this is one one one one. This is not one zero one one It's a big one and then I will somehow create a waveform that here at this point has a higher value for example we May want we don't have to we may want to decide that if there is a problem with the way the bits are received We can correct for this. This is called error correction and we will explain a bit of our error correction It is very important and another thing that we should know all communication channels are using error corrections all of them because it's so powerful and You know that you're using whatever you're doing has error correction even USB Transfer with the hard drive has on the bits that are on the cable error correction everywhere all the time So yes error correction.

So we talked about transmission we talked about Okay, okay, it's time to flash the topic map I think that will be the most This is our course this this picture is the course we're going to teach you everything here Everything here. We're going to be discussing in this course in the I'm not mistaken in the page of the course on canvas. There are also letters or maybe in the reader They're letters and the letters adjust to the modules in canvas so you can say okay Topic number one sampling is covered in module a It's very easy to understand where to find what information So indeed we need to sample Somebody said we need to quantize. We'll talk about quantization one session.

That is the art of converting these samples Into binary digits. This is this We need to do something about signaling. We'll talk about coding how we use Different way of coding and that's important because different codes will produce different use of the spectrum here. This is a ref We'll do some modulation.

There will be some additional optional error error correction applied here somewhere Then we'll send the information through the airways or through an optical fiber It is possible That due to the way we are sending them not only errors will be created or noise will be added But also the waveform will be distorted as is trying to illustrate by this we'll talk about how we can mitigate distortions Due to limited channel performance The signals then will arrive here They'll be decoded the errors will be removed and we will reconstruct the original waveform to create the original message All of this in seven weeks It's a challenge.

We'll manage don't worry Okay course map that was quick Okay signals signals that we're going to use Communication is done in time. So we're gonna say you're gonna talk about time dependent signals The bits I talked about a minute ago also this waveform is a time dependent signals one of the strongest Tools we have in our in our engineering toolbox is the use of frequency in time Whenever we want to do to switch between those this is this frequency time duality But in order to start there, we need to understand what happened. What are signals? So you learn algebra? I think linear algebra you learn about dimensions and orthogonality assume a bit Okay if I want to To present any point at three-dimensional space, you know very well I can look at the Projection of this vector on the three axes XY and Z by breaking down the vector into its different components it makes me it makes possible for me to write the Representation of the number in a simple way. I will say this is one times the unit vector One zero zero plus one times unit vector plus one time and this will give me the results now if for whatever reason The vector I'm trying to I'm trying to analyze the vector that I am observing in space has no Z component So it's in plane here.

I'll take a different color Sorry, then we know that this component is gonna be zero How do we know this we do what is called scalar multiplication?

So we'll multiply the vector this vector With this vector and we ask what the projection and we can calculate the actual component the actual center of the component in this direction This is how we do this how we this is how we process Vectors in N-dimensional space, of course, we cannot draw more than three dimensions. So I'm leaving it at three and The idea behind what we're gonna do after the break is to look for a way to take any periodic function And you've done some with the signature So this is repetition for you any periodic function and write it as a sum of orthogonal vectors Functions are external functions periodic functions And the way the reason we're doing this is because you really really are interested in which frequencies are Inside these signals because the frequencies are important for analyzing the signals, but they will do it after the break. Now, it's your break Hello Grab a seat.

It's a good chalk Let's let's start because they are sitting down.

All right, please take a seat.

We want to continue It's a sunny day outside you want to be outside as quickly as you can so See there works immediately Mark, yes, sorry.

Yes, please and Feedback is valuable.

So we got the feedback that you are not able to open everything that is in the modules now I had a quick check. So what is the issue is that some of the files are in the file system? Locked I Opened a few of them and I need to be carefully going through them because I cannot just open all the files up that we Have within our canvas page then go set the exam for this year. That's what you mean. No, it's just so overwhelming It's giving him piece by piece on everything at once. Okay, and so I will do that So sorry if there's still some some difficulties and I got a question about if there is even more Material to read if you want to go deeper in the in the knowledge And if there's a course book or a book that we're referring to. Yes, there is a book that is in the slide. So There is this course introduction and general information presentation and there's a reader This one. Yeah. Yeah, that's the read as well. But if in this slide set you can also find the book that We once used to build the course on or to use as a reference or whatever. No, you don't need to that's really that's for the One percent or the I don't know the zero point one percent of student that really want to go the extra mile You can find information on the book in this presentation.

Yeah, and I Somebody asked me how to use so I think it's a good idea to ask how to use the mouse Oh here So this is an example of this week's module So this is Our strategy of making it clear, but if it's not clear for whatever reason Share with us what you need and we can adjust this is not so difficult You have the date and you have what's going to be discussed if you click on for yes Yes, which is so you'll get to the page where it is. There's more information on that then for preparation There's an extended quiz on DB and the practice is on Fourier transform an idea something You know the idea something is going to be discussed later But you can practice you can practice the use of DB and DB m's which will likely touch upon and then on Thursday when you come The first couple of minutes of the lecture will be dedicated to this Now if we notice that everybody who shows up already has done the quiz we will start immediately We're not going to waste your time, but we will dedicate the first five minutes to doing this quiz Entry-level quiz so you can test how much you remember from last and DB is maybe an eccentric example But in the next lectures if you start if you look at this in the next modules The first quiz in the in the lecture will be on sampling for example Then you'll be asked to answer simple question about sampling to see where you are and we will look at their answers We can do it in canvas on the back, and we have our own way of looking at the answers And we can say okay this question was done very poorly It's good that we again reiterate something about something for example So this is how we're going to use information, and there's there's additional so if you see at the bottom We call it optional to do so three years ago Three years ago two years ago.

We had we had two very dedicated teaching assistants and They went out of their way to create amazing Matlab content for you which illustrates the basic principles behind the topic we discuss We we call these things mini labs, and they're still available There are apps for Matlab, which you can download and Run on your Matlab code which you all have installed in your laptops And They will really take you through the basic Functionality of something so the mini lab number one for example is about Fourier series you get a an interface where you can look at the different harmonics of a signal and Change their strength and try to reconstruct the script a square wave or a triangular wave for example. This is what we are Talking about so this is an example of the mini labs every week. There will be different mini labs again And not part of the you don't have to do them if you want to go deeper You can use them to to do and they'll be probably every week a few more additional practices because over the years we accumulate a lot of canvas quizzes and They are for practicing, so why not give them all of just can practice. It's good for you That was that yes Okay, so we were we were busy with the vector decomposition and some people came to me in the break and said yeah, but yeah What's what's the use of vector decomposition? This is this is linear algebra We are here in the communications course, and we want to understand how to do time domain changes and whatever So it's not very useful and the answer is no it's not very useful But it's an idea that needs to land and the idea is that you can take one Vector and represent it as a combination of other vectors It is the same way that this You can imagine Without using too much of your imagination that this wave in time Is also a vector of? A lot of points when we sample we take some of them, but in essence a vector is a lot of values and You know this because if you use MATLAB you know that if you want to draw this curve You will have two vectors one will be the values and one of the time series or something So you draw if you want to draw this curve in MATLAB? You will have to insert two vectors one with discrete values of these points and one for the time point It's an XY plot So this is something that you know very very familiar with and the idea behind decomposition of signals And that's where we're working towards is to think about how we can take a signal in this case You want to take a periodic signal? That's where we all we all start so piece of board yes Yeah signals the question is is there a way to Write a periodic signal as a combination of other functions And the answer is yes, and you know the answer for that because you've done the course signals You know you can use cosine functions for example to write this and you've done this before so you know I can say okay, I can add cosine functions and I'm telling you something you know right Look some of you look confused, but you all should know this Okay, illustration if I want to draw cosine function.

I will take I'll take this cosine function and I Will add to it this cosine function Probably I missed already some of these waves But I will add these cosine things to get and if I do this correctly And I make sure the amplitude of each one is cosine is correct. I will get The wave I started with which is a square wave How do I know how much how much of each cosine function I need? It's very relevant question how much to use how much energy because what you get from this is you say okay There is so much orange needed Maybe there's slightly less blue needed, and I'll go on with colors each one of them is there Not all of them are there by the way It's important to realize that not all components are there But each one is there and when they are there they they constitute the wave and how much to use I need to do what I did here I Did projection I took? The vector that was my base vector So in this example somebody asked me in the break what to do. I said okay If my base vector for the z is 0 0 1 and my vector here is 1 2 0 I will multiply these two and I'll find the result This is column multiplication, and I add over all this It's the same with with functions if I want to know how much of this cosine function is In my function. I will take we'll call this function this one This will be f of t I'll take f of t I Multiplied by cosine of a certain frequency. We'll talk about which frequency that is I'll integrate of it over t over 2 t over 2 DT and This this the value of this will tell me how much of this omega zero frequency is Actually in ft because if I multiply cosines omega zero t with ft and the answer is zero if I do that It means that in order to write ft.

I don't need cosines omega zero t. I don't need it It's not there same as when I multiply the vector 1 2 0 by 0 0 1 I get 0 means There's nothing in the z direction It's exactly the same. That's why I draw this picture all the time because it it illustrates very clearly What is the what's the idea of a projection? We project here a vector on the base three vectors of the XYZ Okay, here we project a function on a base function and we ask is there any thing in ft? Which looks like cosine omega zero t the answer is no if the answer to the things right zero It means that ft does not include cosine omega zero t Is that clear? It's very fundamental very basic idea about projection very basic idea about how you reconstruct waveforms, so we need To look for all of these Components we call them Fourier series coefficients and in the in the textbook we We explain in details how we calculate them, and I have even here somewhere Calculation it's all in textbook.

We call these values Ck special for them check and then We do this and instead of having your omega zero we do two pi and ft or K because we need to introduce the Okay One over T, and if you catch me making mistakes in my notations, please let me know I'll write it in the most generic way because it's an expert. It's an order function and this Clearly for every value of K that I choose to To calculate this for Ck will depending on the result of this Integral and the summation and the multiplication will will potentially have a different value Should not surprise you so and what also will not surprise you is that K is just a number here Feels like it's a very complicated thing, but K is just a number so basically I'm taking the function ft and multiplying it by You know this fun.

You know what this you know that he you know these Identities right you know this and for the sign You also know to complete this that this Is equal to cosine? This you know so also here.

There's a cosine and sine hiding inside inside this one, but when you do this multiplication You will get a value for Ck now what is special about this number is Can I choose any value for this? Number this Omega 0 which is a number can I choose any number for it, sorry? Integral No, it could be any number. It doesn't have to be an integral number Sorry Sorry an integer yeah, yeah, I heard you yeah, it doesn't have to be an integer number K is an integer number One two three four five six with omega f is a frequency is a real frequency. What is this frequency? Where is it coming from? What do you think if if this is time and this is Two millisecond three millisecond.

What is the basic frequency of this wave? Yes, so in this case.

What is the period of the signal milliseconds? So what is the frequency? 500 Hertz So for this is it just an example These frequencies base frequency is the Lowest harmonic base frequency of this periodic signal if the period of your signal doesn't matter how it looks It doesn't matter how it looks is One millisecond then it's omega zero this base frequency will be one kilohertz two pi times one kilohertz If your base if your base period is one microsecond Then it will be two times pi times one mega Hertz and This is also a nice thing about time and frequency if you squeeze something in time if you make something happen faster in time It means that the base frequency that is needed to build it is a higher frequency if you want to have higher bitrate if you want to have your mobile phone connect to the Wi-Fi router here and Have 50 megabits per second download speed for Netflix Not sure I'm not sure it's needed, but let's assume it's needed if you want that You need to send information quite fast So the ones and zeros arriving at your phone will be arriving really fast But if they arrive really fast you need a lot of frequency for that Now I don't know how much you know about Wi-Fi routers, but they have channels and frequencies and all sorts of things inside them And you don't and we talk about spectral efficiency later in this course you cannot just Say we're gonna use the frequency as we want.

Oh, something's burning a fire brigade Yeah, that's a joke Hopefully, I'll do the term is burning You will tell us, huh? Okay, very good.

Okay, we're safe the rest can burn so if If my my pulses are faster if I want this information faster shorter pulses because I want the information faster I will need more frequency There is not Infinite frequency I can use it's for some application It's actually something you pay for and some application something that it's just difficult to generate so sometimes difficult but these base frequency this and of course every in principle every harmonic of that frequency can potentially be part of the way I Reconstruct the signal not always not for all signals.

They have different symmetries. For example, this wave as I drew it here And this is the zero is an anti symmetric function It's a function.

It means that if I move the Zero line to here.

It looks like it looks almost like a sine function because it has This behavior Around this this line, which is just a fixed value It really looks like a sine function It does mean That if I try to build the series the composition of this I will not find any cosine function in there in there in there in the series Because the cosine function is a symmetric function But this function is absolutely not symmetric.

In fact, it's anti symmetric. So I will not find cosine functions here I will only find sine functions. For example, just an example The nature of the function we're using to write that we decomposition determines eventually how these signals look so that's really Important to mention that was this What happens when we look at periodic signals and we try to to To decompose them As I said, you will calculate these coefficients these coefficients Will calculate these coefficients and we will see And when we calculate these coefficients We get K so the different values of K.

So the K is one K is two K is three Okay, we'll see for the case of the square wave that we find that there are different coefficients There's something here and something here. There's nothing here. By the way, there's something here Depending on the width of the pulse there is there there's not something here then there's nothing here here something This is symmetric also Of course for every function Every harmonic you have in this you need as I wrote here a positive and negative Euler function so it'll be symmetrical and if you look very carefully you'll see that there is an There's an envelope here and that envelope is a function that you will learn to love in this course It's called the sing function Which is sine Pi x divided by pi x and you learn Calculus and you know you are not allowed to divide by 0 by 0 and this is at 0 you know this but You also learn calculus, you know You can replace sine x with the Taylor expansion of sine x and it has x plus x to power 3 plus x to power 5 Which means that if you divide by x you're left with 1 so sine x over x is 1 for x equals 0 And this is this value here and the rest can be calculated from this formula quite easily and there's a K here to count for the increasing values of x so these and these Components and you can use the mini lab if you want after the lecture you can play with this these components if you add them with the right intensity You reconstruct a sinc function as a square wave. It will just happen almost like magic But what we see here, and that's very important this K space whatever we call it this Discrete frequency space because every one of these Coefficient represent the frequency right this is this K equals 1 is Omega 0 K equals 3 is 3 Omega 0 and so on so each one of these K values represents a frequency which is or is not present in the original function that we're trying to reconstruct We see that because the function in time is periodic The frequencies we need in order to rebuild it are discrete And we'll come to it in a minute so again There's nothing.

I don't need anything in the space between K is equal 1 the K equals 0 In terms of frequency nothing happens here.

There's a frequency that will help me to reconstruct the periodic function there's no power needed nothing it's empty and Same doesn't happen between 1 and 3 nothing here is needed I only need a bit of first harmonic a bit of third harmonic a bit of fifth harmonic to reconstruct the period the signal and this is what we call the Discretization of the spectrum when you talk about periodic signals in time Their spectrum the way they are represented in the spectral domain is built of discrete harmonics Which are all based on the original frequency of the original signal the periodicity of the signal What happens if my signal is not periodic? Can happen? Right, these are all periodic signals What happens my signal is not periodic we can imagine In a way, and this is the way we arrive at this that this is periodic in infinity So it's a bit of a stretch, but you can work with me on this So if we say the period of this is infinity if we say that Yeah if period goes to infinity the frequency goes to zero, thank you It means if before in order to draw our spectrum this drawing We only need to take the base frequency and the third version and the fifth or whatever We still need to do this But the base frequency is zero or as close as possible to zero So the increment in which we are going to draw this spectrum is actually going to be Infinitesimal hence.

It's a continuous function. So if this Has an infinite period and we want to know what is the spectrum of the signal we do the Fourier transform So we write Fourier transform of g of t That will be in from minus infinity to infinity Note there is no omega zero anymore It's no longer a unique frequency.

There's no K anymore It's just omega right, it's not specific and for every value of omega every value of omega you can calculate the integral and give a number So for every value longer I can do I can put the whole function in Choose omega is one point thirty six Hertz Calculate this and get a number and I'll get a number and I will draw a point and I can do it again As much as God, but I can use math which is what we do You can calculate and for many many functions you can calculate directly what the Fourier transform is of gt Not for all functions, but for many functions That's what you're going to do when you try to ask yourself. What is the fun? What is the free transform of this? Why do we care about this? Why is it so important because you learned? signals and You learn about transfer functions you did right? H of t you know h of t.

Yeah, there's input and there's output This is all known stuff.

How do I know what is the output if I know the input and the transfer function?

Yes Convolution Call this oh and this I So all of t is equal to the convolution between IFT and HD convolution nasty nasty function You don't want to do convolution What do you do used for yay? For yay is very kind to us because it's okay if I Can do convolution as I said it's not nice. Maybe now with chat GPT. It's easier, but it used to be very annoying I can also say if I know how the spectrum of the input looks like now the spectrum and what the Transfer function looks like in the frequency domain. I can replace convolution by multiplication That's the power Fourier transform, and I'm not going to prove any of this, but this is in time and in frequency oh Of f is equal to I of f Multiply by h of f life is good So much better example, huh h of t here.

What is this? Lopez filter, thank you I Always confuse them, so I watched my video of last year to memorize that it's a lopez filter.

This is a lopez filter you can Calculate the time dependent response of this filter.

It's an exponential decay You've learned this I hope in circuits, and this no not really okay. You should have but okay It's the first order differential equation solution with a dish condition doesn't matter you can calculate this and this this will have Something like this will look like this in time h of t So you charge it and then the discharges slowly or the other way around doesn't really matter In the frequency domain you know this somebody said lopez filter This is time in frequency it's something like this This is the cutoff frequency.

This is h of f Sorry h of f Function like this this this is you know, and this is very easy now. You can say okay. I put in for example 100 Hertz signal in I Look at h of f it tells me that at 100 Hertz the transmission is Perfect, or I lose 3 DB or I lose 10% or whatever this graph is very well And then I can say how much of 100 is left the output Multiplication the powerful application so we use Fourier transform a Lot in this course because we want to know We really want to know what spectrum we are capturing how much of the spectrum we are using and again it's not different than time series this this this oscillating square wave if Phenomenon time are taking less time if it's shorter you need more spectrum. This is this always works. It's kind of a given a given thing and Again the the mind step you need to do from the series to the transform is just say it's an infinite Period hence the frequency is zero the base frequency is zero So actually calculating it for every value of the number and we generalize Omega So we when we solve this It will be a function of s And again, the example we always use the same example.

As I said, we need to learn to love distinct function is This single pulse in time.

This is for yay transform This way and you get Sync function and this one over t minus one of the width the width of this pulse Inversely proportional to the width of this hence if this is double the width slower Then the frequency will be narrower Slower means narrower frequency if I shrink this pulse and make it shorter in time.

This will double and more frequency And this is all in the reader. You can read about it. You can find all the information Okay, there are a few Few not a lot a few important Fuye Transforms that you need to know or you should know or you should care about to memorize These are kind of the basic function. So the most Obvious one would be and you we've done this already is what is the Fuye transform of? cosine You don't know we do it in steps first step We're gonna do is we're gonna ask ourselves. What is the Fuye transform? Of me of this and you can use the function there to substitute.

So I write here Fuye of so far Maybe it is a factor one over something. I don't care about the fact that is a constant Is it possible to calculate this integral can I calculate? What is the integrant here before? from integral from integral theory You agree with this step I did so the K I added the two exponentials together and Then one more step is to actually write what it is. This is one over J Omega 0 minus Omega Because an exponential in you integrate away explanation you get an exponential Yeah, if I'm making crude math errors, please correct me.

I'm doing it Can you calculate this number? What does it mean to substitute T is infinity? Into this and into this Minds infinity. Can you actually tell me what is the value of this function infinity? Yeah, no, no, it's impossible because the nice thing about infinity. You can always add a bit more Yeah, that's the nice thing about infinity. So you say okay for that value of infinity. It's zero if I had a bit more infinity It's something else.

So yeah, you cannot do this. I agree There's only one case where this thing collapses, of course When the T doesn't matter anymore and that's when the Omega is equal to Omega 0 see it If Omega is equal to Omega 0 I have here 1 divided by 0 not good multiply by 1 Because this whole thing is becomes the exponent power 0 that's 1 and I divide by 0 Not good we get infinity We get infinity as the result of this integral if The frequency at which I'm trying to calculate it is equal to Omega 0 How do I mathematically draw it?

I? Look at them Scale say this is Omega 0 and there are an arrow I Call this a Delta function and I say that at that point there's infinite energy with zero frequency Okay, that's a Delta function.

It's the result of the fact that when I try to calculate this integral I will not be able to do it except for when Omega's Omega equals Omega 0 Okay, remember that cosine function Has two of these guys One in the plus and one the minus Right, I can write this it's written over there on the board Cosine Omega 0 t is exponent plus plus divided by 2 So I can write that the Fourier of this Is the Fourier of this I get two Delta functions One for the plus frequency one for the minus frequency. Why is this important? Because we will talk later in this course about how we send information and when we send information We what we'd never do in communication is we send information. What's called baseband We never send the information the frequency is generated. We always have a carry. You know that Wi-Fi is 2.4 gigahertz You know that the 2.4 gigahertz has nothing to do with the information you're sending the speed at which it's and it's just a radio wave which oscillates at 2.4 gigahertz and We superimpose we modulate it We'll talk about modulation this course later with the information want to send We will show that the modulation is nothing more nothing less than multiplying Your function in the time domain with a cosine function which will Result in a convolution in the frequency domain which will result yes, please.

It's true It doesn't really matter and yet when we talk about what happens to signals when we multiply by cosine function It's important to distinguish that half of the power goes To the hub higher frequency and the other half goes to the lower frequencies So effectively because it's a hub about preservation of energy at the end here where this Yeah Yeah, I can Would it help me? Negative Shay is just square root of minus.

I'm lost Okay, okay, we'll have a break in five minutes, okay How will the spectrum for the sine function look like now it's now it's using the knowledge of gain I know it's a long afternoon, but try The function is written over there So within if I want to find a Fourier transform in the sine function, what will it look like? Yes minus, okay.

Yes. Thank you. Yeah for yay Sine will be one over 2j Delta and maybe I've swapped the minus plus.

Okay questions Properties For you transform has a lot of useful properties.

They're all Listed in the reader. There's a table, but you can find them. Of course online whenever you want them Fourier transform is a linear operation which means that Fourier of X A X plus B Y Is equal to a X of F plus B Y of F That's a linear operation It has Which properties I want to focus on it has so this is a very important one it has The property of time translation a frequency solution if you multiply a function with an Exponential which is just this is just linear phase If you draw this in the phase you see this is just linear phase as T as time This is a constant as time goes up this phase Increases so it's a linear phase So if you multiply your f of t with this kind of linear phase you see that it produces a frequency shift For the expression the frequency will shift and if you have a time shift In For the function.

So if you want to to ask yourself, what is the f of X of t minus t 0 then it will be X of f With an additional Phase shift because this is a fixed number you can think about what it happens if I put this into the integral It's will become a fixed number that just goes out of the integral Because if G includes e to the j omega 0 t and you can just take it out of the interval So yeah, oh sorry What other properties which are relevant?

I think these are the major ones Yeah Yes, please.

Yeah Before everybody is said jumping to the break. I have one smaller announcement in the last break someone lost something really valuable I Have it here If the person can explain to me what that person lost Please let me know and I give it back, but it's so valuable that I need a clear description of that. So Thanks for understanding We'll take a break now. It's a good that you say not say what it is